Roll No. ....

TEE–301(ECA)

B. TECH. (EE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ELECTRICAL CIRCUIT ANALYSIS

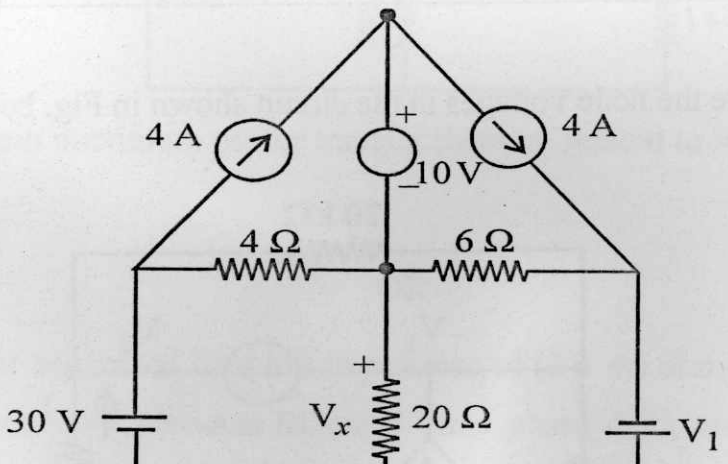
Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

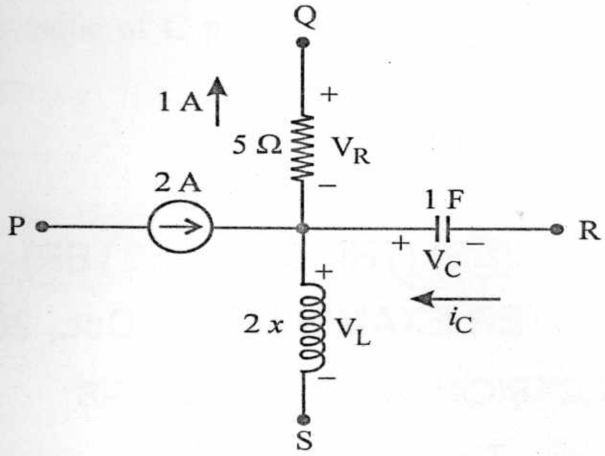
(ii) Each sub-question carries 10 marks.

1. (a) Using nodal analysis in the circuit shown in Fig. below, determine what value of V_1 cause V_x is equal to zero. (CO1, CO2)

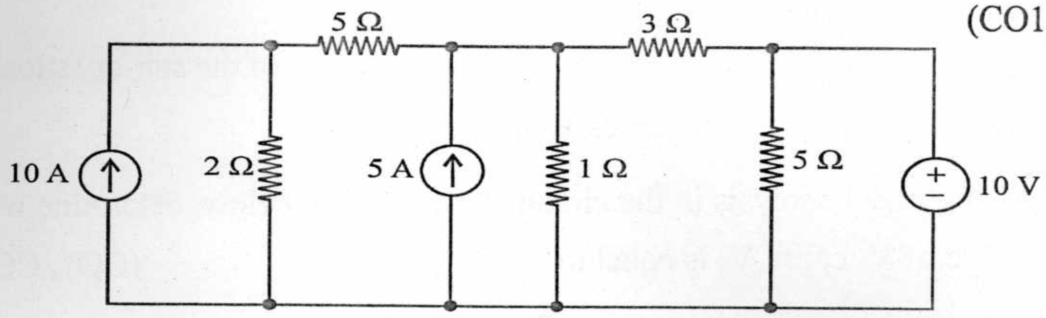


P. T. O.

(b) A segment of a circuit is shown in figure below. $V_R = 5$ V, $V_C = 4 \sin 2t$. Find the voltage across the inductor. (CO1, CO2)

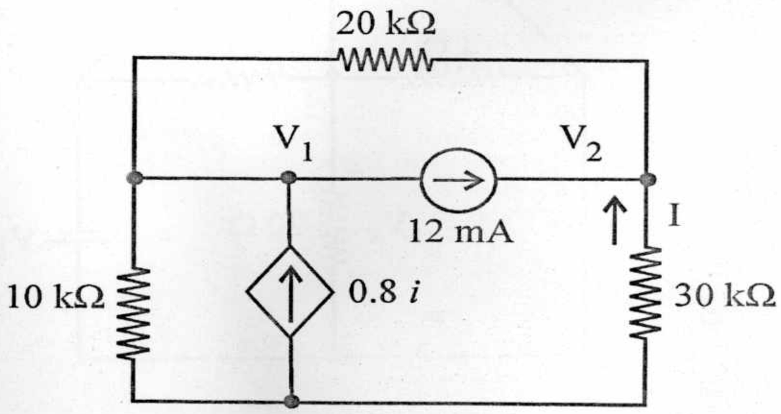


2. (a) Find the current in the 3 Ω resistor for the circuit shown in Fig. below : (CO1, CO2)

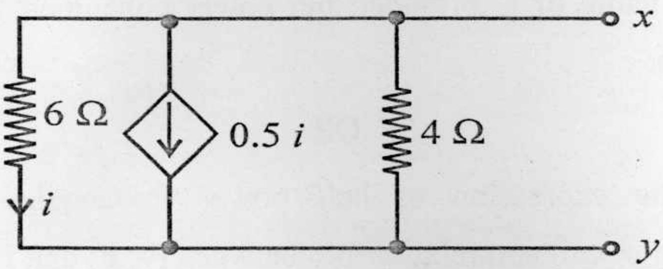


OR

(b) Calculate the node voltages in the circuit shown in Fig. below : (CO1, CO2)

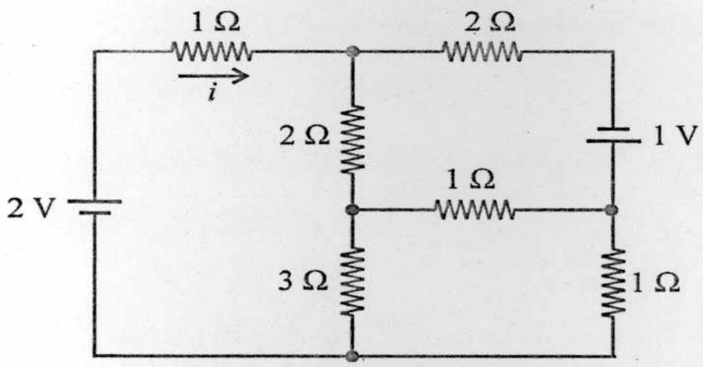


3. (a) Find the Norton's equivalent circuit to the left of the terminal x-y as shown in Fig. below : (CO1, CO2)



OR

(b) Find the network shown in the figure, draw the oriented graph and obtain the tie set matrix. Use this matrix to find current i. (CO1, CO2)



4. (a) Explain maximum power transfer theorem related to AC circuits. (CO1, CO3)

OR

(b) A star connected load has impedance of $(3 + 4i)$ ohm in each phase and is connected across a balanced three phase delta connected alternator having line voltage of 120 V. Obtain the line current of both the load and generator. (CO1, CO3)

5. (a) A series resonating circuit has a source frequency of 5 kHz and source impedance of $(2 + 4i)$ ohm. The load impedance being $(10 - iX_c)$ ohm. Find the value of C provided the power consumed by the resistor is maximum. (CO1, CO3)

OR

(b) Derive the expression of half power frequencies in series RLC resonating circuit and relationship between F_0, F_1 and F_2. (CO1, CO3)